Controls And Data Protocol

Rule

A force reading is not interpretable until the test article has produced a self-read receipt. The order is:

1. prove the active device has repeatable drive/read behavior,
2. prove `ACTIVE_PLUS` and `ACTIVE_MINUS` produce opposite signed internal contrast,
3. measure force with the device running from onboard power,
4. reject ordinary artifacts with sham, dummy, flip, thermal, electrostatic, magnetic, and vibration controls.

Required States

State	Role
`OFF`	Baseline drift and environmental record.
`SHAM`	Same average power, heat, RF envelope, and timing as active mode, without the coherent self-read branch.
`ACTIVE_PLUS`	Coherent state with predicted positive sign along the declared force axis.
`ACTIVE_MINUS`	Same power and timing, reversed sign along the declared force axis.
`FLIP`	Same physical device rotated or inverted so the internal command faces the opposite lab direction.
`DUMMY`	Matched artifact witness without a valid coherent self-read branch.
`HEATER_ONLY`	Same thermal envelope without resonant drive.

Self-Read Receipt

Before any force claim, save:

- port layout and photos,
- firmware hash,
- state schedule,
- frequency sweep,
- coupling matrix,
- ringdown or phase features,
- repeatability plot,
- shuffled-record prediction control,
- top/bottom or A/B signed contrast,
- dummy comparison.

Minimum pass:

- coupling matrix repeats within preregistered tolerance,
- held-out readouts are predicted better than shuffled records,
- signed contrast reverses between `ACTIVE_PLUS` and `ACTIVE_MINUS`,
- dummy lacks the same signed contrast.

Pendulum Protocol

Use for the piezo-crystal plate.

1. Measure moving mass `m`.
2. Measure pendulum length `L`.
3. Let the device thermally settle.
4. Record camera or laser baseline in `OFF`.
5. Run `SHAM`, `ACTIVE_PLUS`, `ACTIVE_PLUS`, `SHAM`.
6. Run `SHAM`, `ACTIVE_MINUS`, `ACTIVE_MINUS`, `SHAM`.

7. Rotate the device 180 degrees and repeat.
8. Run the dummy with the same schedule.
9. Run heater-only or resistor-only at the same average power.

Direct camera force:

```
F = m * g * x / L
```

Laser mirror force:

```
F = m * g * s / (2 * D)
```

where `x` is bob displacement, `s` is laser spot displacement, and `D` is mirror-to-screen distance.

Balance Protocol

Use for the plate if a 0.1 mg or 1 mg balance is available.

1. Put the balance on stone or an anti-vibration table.
2. Close the draft shield.
3. Calibrate with known masses.
4. Run only battery-powered devices on the pan. No USB, power, or ground cable.
5. Use same-object ABBA blocks:

```
SHAM_1, ACTIVE_PLUS_1, ACTIVE_PLUS_2, SHAM_2
SHAM_1, ACTIVE_MINUS_1, ACTIVE_MINUS_2, SHAM_2
```

6. Discard the preregistered settling interval from every state.
7. Save raw balance CSV, not only screenshots.

ABBA estimator:

```
delta_m_X = mean(X_1, X_2) - mean(SHAM_1, SHAM_2)
F_lift = -g * delta_m_X
```

Acoustic Force Protocol

Use for the cymbal or dish rigs.

1. Keep the driven dish support independent of the ground-plate load cell.
2. Sweep standoff: 0.5, 1, 2, 5, 10, 20, 30 mm.
3. Sweep frequency around resonances.
4. Sweep drive amplitude.
5. Repeat on reflective and lossy surfaces.
6. Record temperature and acceleration at each dish.
7. Repeat `ACTIVE_PLUS`, `ACTIVE_MINUS`, and `SHAM` at matched power.

Conventional acoustic force should change with standoff, surface, resonance, and drive amplitude. A residual that ignores those variables needs extra scrutiny, then flip and dummy tests.

Artifact Table

Artifact	Mechanism	Required control
Air currents	heating drives convection	closed shield, thermal settling, heater-only run
Buoyancy	density change near device	temperature log, matched dummy, regression against temperature
Thermal radiation	asymmetric heating of pan or shield	emissivity match, IR check, same-power sham
Electrostatics	charged surfaces pull on pan or frame	discharge protocol, Faraday shield, handling log
Magnetics	currents or steel parts couple to environment	nonmagnetic fixtures, magnetometer log, rotation test

Artifact	Mechanism	Required control
Cable forces	wires act as springs	onboard battery and logging, no cables during force run
Acoustic leakage	sound pushes air or nearby surfaces	enclosure, microphone/accelerometer log, standoff map
Vibration	mechanical coupling into supports	anti-vibration table, accelerometer log, dummy
RF/EMI	drive electronics affect sensors	shielding, dummy electronics, spectrum check if available
Center of mass	moving parts or wires shift load	solid-state switching, no relays, pan-position test
Human handling	order and expectation bias	preregistered schedule, blind labels where possible

Decision Rules

Development signal:

- a repeatable force or deflection appears in one configuration.

Candidate residual:

- signal tracks the measured self-read scalar,
- `ACTIVE_MINUS` reverses sign,
- physical flip reverses sign in lab coordinates,
- dummy and sham reject the signal,
- temperature, battery, acoustic leakage, vibration, electrostatic, and magnetic logs do not explain it.

Publishable upper bound:

- no candidate residual,
- force sensitivity and artifact controls are documented,
- raw logs and analysis scripts are included,
- the null is stated as a bound for that hardware, not as a universal disproof.

Evidence Bundle

Save one directory per run:

```
run_id/
  manifest.yaml
  firmware/
  photos/
  raw/
    balance.csv
    pendulum_position.csv
    load_cell.csv
    device_log.csv
    environment.csv
  analysis/
    analysis_script.py
    frozen_config.yaml
    plots/
    summary.md
  hashes.txt
```

The result lives in the bundle. A video without the bundle is outreach, not evidence.